

Behavior of Mg and C-O isotopes during mafic magma-carbonate interaction at the Jinchuan Ni-Cu deposit, North China Craton

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ABSTRACT

The Jinchuan Ni-Cu deposit is one of the largest magmatic sulfide deposits in the world, but the cause of sulfide saturation remains unclear. It has been suggested that sulfide saturation in the Jinchuan magma was triggered by oxidation through assimilation of fluids related to de-carbonation of marbles. This study uses Mg and C-O isotopes to evaluate the process of carbonate assimilation. The measured values of $\delta^{18}\text{O}$ and $\delta^{13}\text{C}$ for carbonate minerals from country rock marbles range from 13‰ to 22‰ and from −1.7‰ to 1.3‰, respectively. Carbonate minerals from xenoliths and hybrid rocks in the contact zone have lower C-O isotope ratios, with $\delta^{18}\text{O}$ from 11‰ to 16‰ and $\delta^{13}\text{C}$ from −4.7‰ to −2.3‰, respectively. The carbonate $\delta^{26}\text{Mg}$ values of xenoliths and hybrid rocks are from −1.71‰ to −1.36‰, similar to the values of the marbles (−1.74‰ to −1.34‰). The whole-rock $\delta^{26}\text{Mg}$ values of hybrid rocks samples associated with xenoliths vary from −1.69‰ to −0.92‰. The $\delta^{18}\text{O}$ values of pyroxene at the margin and the center of the Jinchuan intrusion vary from 5.1‰ to 7.8‰ and from 5.3‰ to 6.5‰, respectively. Despite varying distances from contacts with country rocks, pyroxene in these samples are characterized by $\delta^{26}\text{Mg}$ values from −0.32‰ to −0.16‰, which are similar to the typical value of the Earth's mantle (−0.25‰ $\pm$ 0.07‰).

The C-O isotopic variations are consistent with de-carbonation. Rayleigh fractionation models suggest from 40% to 80% CO_2 loss from xenoliths and hybrid rocks. The Mg isotopic compositions in the contact zone reflect reactive assimilation between xenoliths and mafic magma. Product olivine and diopside inherited the $\delta^{26}\text{Mg}$ value of reactant dolomite. The transport of released CO_2 from xenoliths into magma is consistent with increasing oxygen fugacity and potentially triggering sulfide saturation. However, Mg isotope exchange was confined to the contact zone, and appears to have been unrelated to the ore-forming process.

1. Introduction

The formation of a magmatic Ni-Cu-PGE deposit is strongly linked to the separation of a metal-rich immiscible sulfide liquid from a mafic to ultramafic magma. There are several possible processes that can cause a magma to become sulfide saturated. These include fractional crystallization, addition of externally derived sulfur, and changes in melt composition and/or intensive variables related to assimilation of country rocks and contamination (e.g., Ripley and Li, 2013). For many deposits the assimilation of sulfide or sulfate is a principal process that can be verified both by the presence of S-rich country rocks and xenoliths, and anomalous (non-mantle) sulfur isotope ratios in the ores (e.g. Leshner

and Keays, 2002; Robertson et al., 2015). The Jinchuan deposit in China is one of the world's largest magmatic Ni-Cu deposits, but is one where strong isotopic evidence for external S addition has not yet been found. As an alternative to external S addition, Lehmann et al. (2007) proposed that CO_2 -rich fluids derived via assimilation and de-carbonation of country rock dolomitic marbles oxidized the magma. The oxidation led to a reduction in the capacity of the magma to dissolve sulfide, and sulfide liquid saturation resulted. Tang et al. (2018) reported $\delta^{26}\text{Mg}$ values for olivine and pyroxene separates and for whole rocks of the ultramafic intrusive rocks ranging from −0.40‰ to −0.26‰, and inferred the addition of <10% of marble to the Jinchuan magma. Their weighted mean $\delta^{26}\text{Mg}$ value of −0.31‰ $\pm$ 0.06‰ for the silicate

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minerals is indistinguishable from the value of the mantle ($-0.25\text{‰} \pm 0.07\text{‰}$; Teng et al., 2010).

In order to further test the hypothesis that interaction with carbonate rocks was important for ore genesis, we have conducted C and O isotope studies of country rock marbles at Jinchuan, and Mg isotope studies of both marbles and sulfide-bearing igneous rocks. This study focuses on samples collected from the contact zone between country rock marble and the Jinchuan intrusion. Petrographic analyses and mineral compositions are utilized to reconstruct chemical reaction paths. Carbon and oxygen isotope ratios of carbonate country rocks are used to constrain the temperature of de-carbonation and to constrain the amount of released CO_2 . Magnesium isotopes are considered to be an effective tool for tracing crustal contamination of magma (Shen et al., 2013; Huang and Xiao, 2016; Ke et al., 2016). In this study Mg isotope ratios of country rocks and the Jinchuan intrusion have been measured to help evaluate the potential assimilation of carbonate rocks by mafic magma. The combined results are used to provide a better understanding of the role of magma-carbonate interaction in magma oxidation and its potential importance to the formation of the Jinchuan magmatic Ni-Cu sulfide deposit.

2. Geological setting

The Jinchuan magmatic Ni-Cu sulfide deposit is located in the eastern part of the Longshoushan terrane at the southern rim of the Alxa block, in the westernmost part of the North China Craton (Fig. 1a). The Longshoushan Terrane is bounded by two regional thrust faults on its northern and southern sides, and the North Qilian Orogenic Belt, which occurs to its south (Fig. 1b). The North Qilian Orogenic Belt was formed by the collision between the Alxa block and the North Qilian block at ~ 450 Ma (Song et al., 2013). In the Longshoushan Terrane, Proterozoic metamorphic rocks are unconformably overlain by Paleozoic conglomerate, sandstone, and limestone. Abundant Paleozoic granites are present in the Longshoushan terrane and within the adjacent North Qilian Orogenic Belt. There are several mafic-ultramafic intrusions in the Longshoushan terrane, all of which except Jinchuan appear to be unmineralized (Tonnelier, 2010). Available zircon U-Pb isotope dating shows two episodes of mafic magmatism in the region. The older one took place at ~ 830 Ma as represented by the Jinchuan ultramafic intrusion (Li et al., 2005; Zhang et al., 2010), and the younger one occurred at ~ 420 Ma as represented by the Xijing mafic intrusion and mafic dykes (Duan et al., 2016).

The Jinchuan ultramafic intrusion intruded Proterozoic marbles and gneisses (Fig. 1c). The current surface exposure of the Jinchuan intrusion is ~ 6000 m long and ~ 300 m wide. The overall strike of the intrusion is northwest and dips are to the southwest with oblique angles varying from 50° to 80° . The Jinchuan intrusion is divided into four segments by a series of northeast-trending strike-slip faults (Fig. 1c). They were numbered as III, I, II and IV from west to east by mine geologists based on the order of mine development. Segment II is exposed and located in the middle portion of the Jinchuan intrusion. Segment I is also exposed to the west of Segment II. The unexposed Segments III and IV occur at the western and eastern terminations of the Jinchuan intrusion, respectively.

The country rocks surrounding the northern portion of the intrusion are composed of marble (Fig. 1d). Many dikes of amphibolite and a few pegmatites and red granites intrude the marble. In the marble bedding is preserved with layers of variable thickness (20 cm to 1 m) that are generally concordant with the northern margin of the Longshoushan belt as well as the northern contact of the intrusion. Marble xenoliths are common along the northern contact and appear sporadically throughout the intrusion. The shape of xenoliths varies from angular to round, and the size of xenoliths can be as large as 60 cm wide and 4 m long (Supplementary Fig. S1). The xenoliths have sharp contacts with the surrounding rocks and appear to be stopped fragments with no evidence of partial melting (Supplementary Fig. S1).

The ultramafic intrusive rocks host ore bodies, and sulfide ore bodies occur over a range of depths from surface down to the maximum ~ 400 m below surface (Fig. 1e). Net-textured sulfide mineralization is the principal ore style; mineralized zones are commonly characterized by a thin margin of disseminated sulfides. Massive sulfide mineralization is rarely present. The most common sulfide minerals are pyrrhotite, chalcopyrite, and pentlandite (Table 2).

3. Petrography

3.1. Marbles

The country rock marbles vary from white to pale green, and are composed of nearly pure dolomite (Fig. 2a), or dolomite with a small amount of calcite. A few samples contain calcite and olivine, with minor clinozoisite, tremolite and hydrogrossular.

3.2. Xenoliths and hybrid rocks in the contact zone

Xenoliths consist of diopside with minor amounts of calcite (Fig. 2b); the original marble has been almost totally de-carbonatized. In the contact zone, xenoliths are present within hybrid rocks. Some hybrid rocks contain isolated olivine and pyroxene grains, which are distributed irregularly in the carbonate-rich matrix (Fig. 2c), and others show large patches of olivine surrounded by the carbonate-rich matrix (Fig. 2d). The matrix is enriched in calcite with only small amounts of dolomite, as well as small grains of olivine and pyroxene (Fig. 2e).

3.3. Intrusive rocks

The ultramafic rocks collected from the margin of the Jinchuan intrusion show little evidence of interaction with country rocks. Mineral assemblages are composed of olivine, pyroxene and plagioclase (Fig. 2f), with no perturbation to textures characterized by primocrysts and interstitial minerals. Sample (JC 27) is anomalous as it contains small dolomite grains, which are not observed in other intrusive samples.

The rock samples from the interior of the Jinchuan intrusion are olivine-orthopyroxene-chromian spinel cumulates. The interstitial material is composed of variable mixtures of orthopyroxene, clinopyroxene, plagioclase, and sulfides.

4. Samples and analytical methods

4.1. Samples

Most of the samples of this study were collected from the surface, with emphasis on the contact relationships between the Jinchuan intrusion and country rocks (Fig. 1c). The petrographic classifications of the samples are listed in Table 1. More detailed information on the regional contact relationships can be found in Lehmann et al. (2007). Nineteen surface samples were analyzed for mineral chemical compositions using an electron microprobe, and for carbon, oxygen, and magnesium isotope ratios.

Underground samples were collected from the interior of the deposit along underground tunnels: 4 samples (II9-2 to II9-5) from II-9 at the 958 m elevation, and 3 samples (II25-3 to II25-9) from II-25 at the 1050 m elevation (Fig. 1e). The tunnel elevations correspond to 700 to 800 m below the surface. These samples were analyzed for oxygen and magnesium isotope ratios.

4.2. Analytical methods

Mineral compositions were determined by wavelength-dispersive X-ray emission spectrometry using a JEOL JXA-8100 electron probe microanalyzer (EMPA) at the Second Institute of Oceanography (SOA), China. The analytical conditions were 20 nA beam current, 15 kV

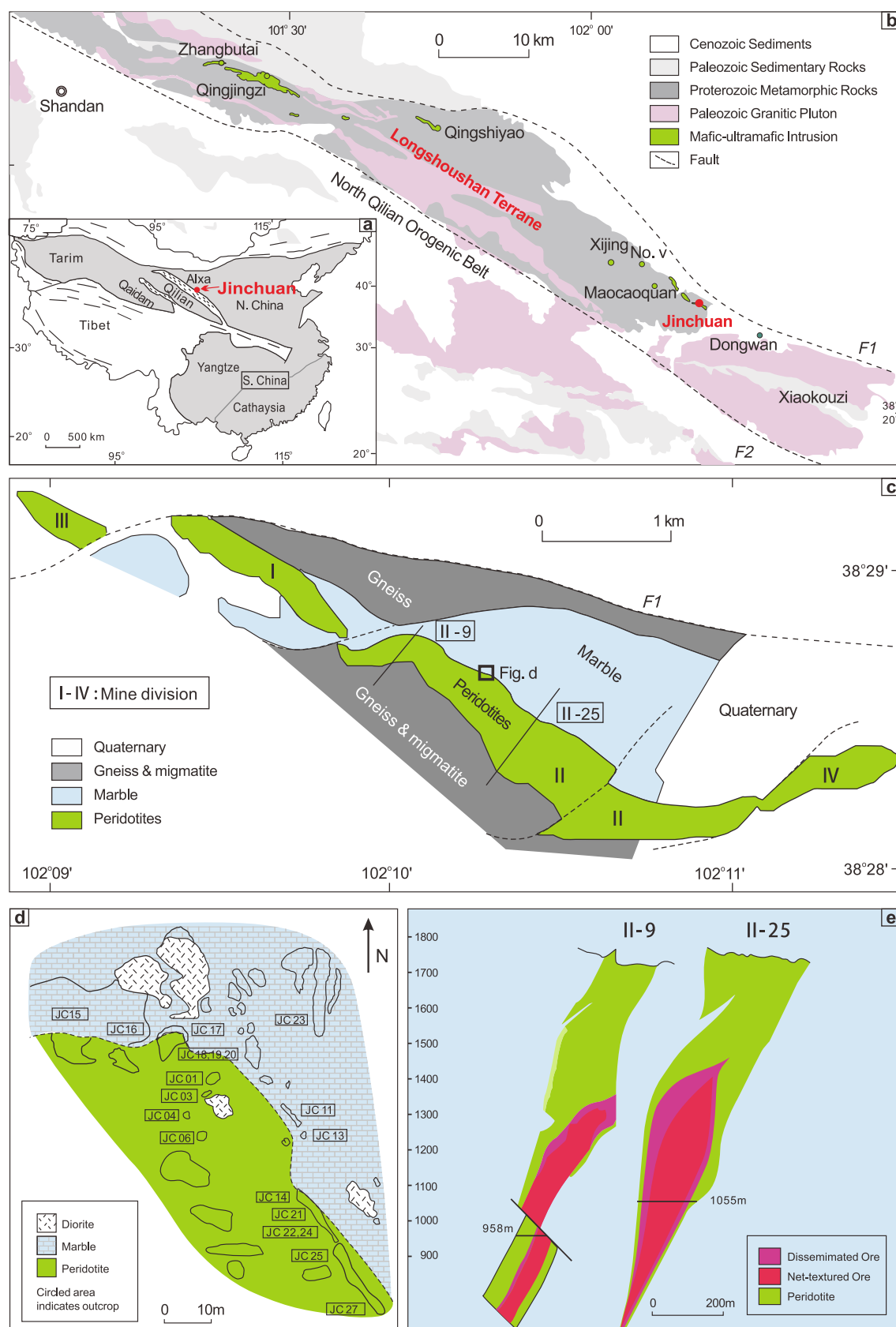


Fig. 1. (a) The locations of the Jinchuan Ni-Cu sulfide deposit in China, (b) the simplified geological map of the Longshoushan terrane, and (c) the geologic map of the Jinchuan intrusion and surrounding rocks. Figures (a)–(c) are after [Li and Ripley \(2011\)](#). (d) Detailed sketch map of part of the northern contact of the Jinchuan intrusion, showing the sampling sites. Figure (d) is modified after [Lehmann et al. \(2007\)](#). (e) The cross sections with sampling sites. Figure (e) is modified after [Duan et al. \(2016\)](#).

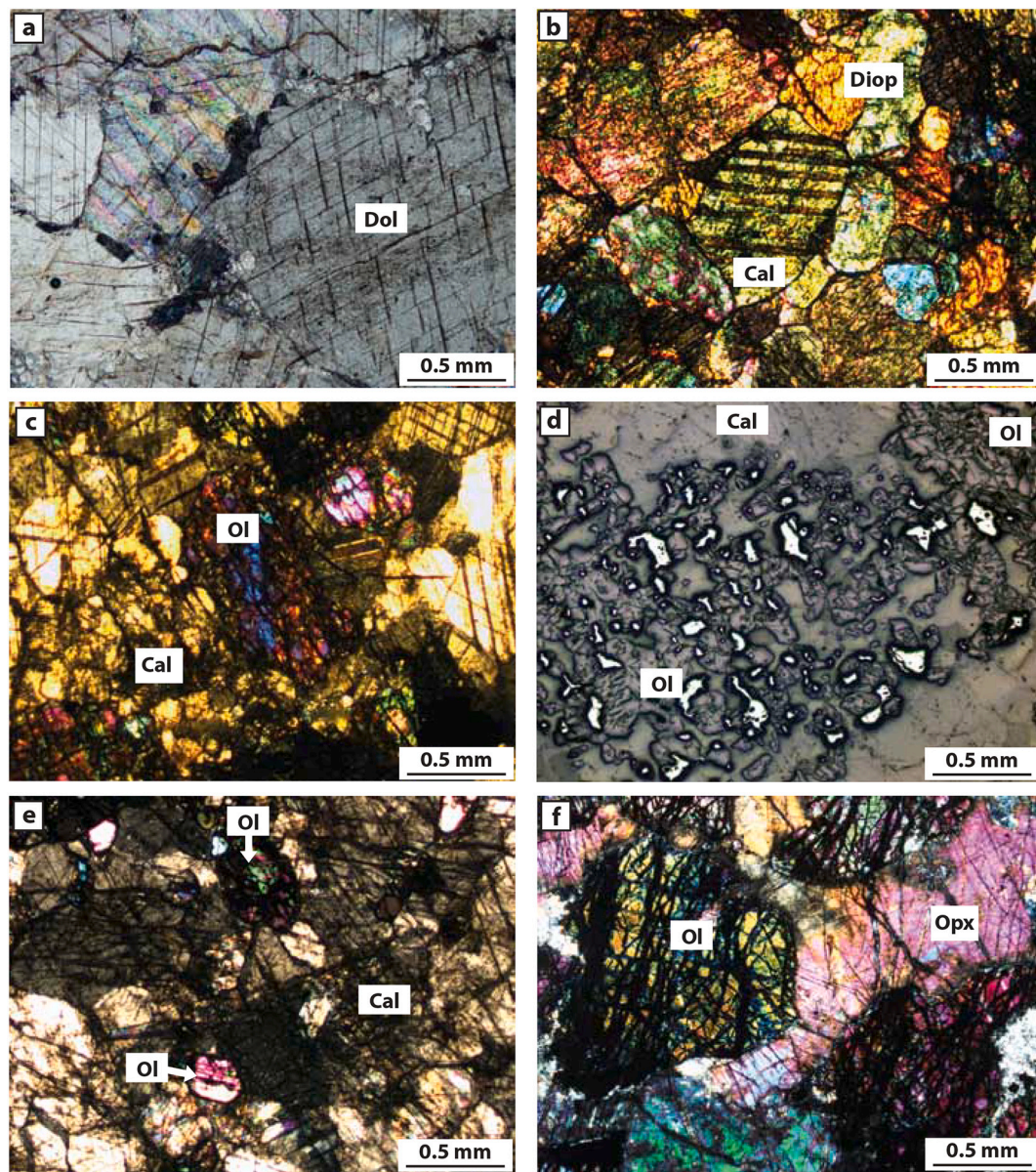


Fig. 2. Photomicrographs of thin sections from country rock, the contact zone and the intrusion. (a) Country rock marble composed of dolomite and calcite. (b) Marble xenolith transformed into diopside with less amount of calcite. (c) Hybrid rock showing isolated olivine grains within in carbonate-rich matrix. (d) Hybrid rock containing patches of irregularly distributed olivine grains under reflected light. (e) The matrix is calcite rich and contains small grains of olivine. (f) The mafic rocks collected from the northern margin of the intrusion, showing olivine cumulate with interstitial orthopyroxene.

acceleration voltage, 1–10 μm diameter beam size, and 20–60 s counting times. Standard minerals (USA SPI Supplies) used in EPMA analyses included the minerals albite, apatite, diopside, olivine, orthoclase, oxides (NiO , TiO_2 and Cr_2O_3), pyrope and rhodonite.

Carbon and oxygen isotope ratios of carbonate mineral separates were determined at the Indiana University Stable Isotope Research Facility. CO_2 from carbonate mineral separates were generated by reacting anhydrous ortho-phosphoric acid in He purged exetainers with the samples at 75 $^\circ\text{C}$ for 1 h. Isotopic compositions of carbon and oxygen were measured on a Thermo Finnigan Delta Plus XP mass spectrometer. Isotope ratios are reported in standard delta notation, with multiplication by 1000 to convert values to per mil. Values are given relative to Pee Dee Belemnite (PDB) for carbon isotopes and Vienna Standard Mean Ocean Water (V-SMOW) for oxygen isotopes.

Oxygen isotope ratios for pyroxene separates were determined at the Indiana University Stable Isotope Research Facility. Oxygen was released using the conventional bromine pentafluoride extraction

technique. Liberated oxygen was reacted with a graphite disk under vacuum, held by a Pt-Rh wire in a heated chamber to produce CO_2 gas. The CO_2 gas was then analyzed on a Thermo Finnigan Delta Plus XP mass spectrometer. The value of the NBS-28 quartz standard in the lab is $9.6 \pm 0.2\text{‰}$ (V-SMOW); instrumental analytical precision was $\pm 0.05\text{‰}$.

Mg isotope ratios for carbonate mineral and pyroxene separates were measured using a Thermo Scientific Neptune Plus multi-collector inductively coupled plasma mass spectrometer following the method of An et al. (2014) at the University of Science and Technology of China. Mineral separates were digested by a 3:1 mixture of 1.5 ml concentrated HF and 0.5 ml concentrated HNO_3 , and then evaporated to dryness. Following digestion by 2 ml aqua-regia and then evaporation to dryness. Finally samples were dissolved in 2 M HNO_3 for chromatographic column chemistry. The Mg purification was then performed in Saville micro-columns loaded with 2 ml of BioRad AG50W-X12 resin. The blanks were around 5 ng, negligible relative to the amount of purified Mg (20–40 μg). The Mg isotope ratios were measured using the standard

Table 1

Petrographic description and C-Mg-O isotope data for Jinchuan surface and underground samples.

Sample	Location	Rock type	$\delta^{13}\text{C}_{(\text{carb})}$ (‰)	$\delta^{18}\text{O}_{(\text{carb})}$ (‰)	$\delta^{26}\text{Mg}_{(\text{carb})}$ (‰)	2SD [#]	$\delta^{18}\text{O}_{(\text{Px})}$ (‰)	$\delta^{26}\text{Mg}_{(\text{Px})}$ (‰)	2SD
JC15	surface	marble	0.79	20	-1.54	0.02			
JC16	surface	marble	1.3	20	-1.60	0.04			
JC17	surface	marble	-1.7	13	-1.51	0.05			
JC18a	surface	contact-xenolith	-3.1	15	-1.48	0.05			
JC18b	surface	contact-hybrid rock	-3.2	11	-1.53	0.03			
JC19	surface	contact-hybrid rock	-4.7	11	-1.53	0.03			
JC20a	surface	contact-xenolith	-4.3	12	-1.48	0.05			
JC20b	surface	contact-hybrid rock	-3.3	15	-1.71	0.04			
JC01	surface	contact-hybrid rock	-3.1	16	-1.52	0.04			
JC03	surface	contact-hybrid rock	-2.3	13	-1.36	0.05			
JC04	surface	intrusive					7.8	-0.20	0.03
JC06	surface	intrusive					6.5	-0.26	0.05
JC23	surface	marble	0.22	22	-1.34	0.04			
JC11	surface	marble	0.12	19	-1.74	0.03			
JC13	surface	marble	-1.4	21	-1.51	0.04			
JC14	surface	intrusive					5.1	-0.21	0.04
JC21	surface	intrusive					5.1	-0.29	0.04
JC22	surface	intrusive					5.2	-0.24	0.05
JC24	surface	intrusive					6.2	-0.16	0.04
JC25	surface	intrusive					6.0	-0.23	0.03
JC27	surface	intrusive	-0.08	16	-1.59	0.06	5.5	-0.23	0.02
II25-3	underground	intrusive					6.1	-0.32	0.00
II25-7	underground	intrusive					5.3	-0.16	0.02
II25-9	underground	intrusive					5.5	-0.17	0.03
II9-1	underground	intrusive					6.0	-0.26	0.03
II9-2	underground	intrusive					6.0	-0.23	0.01
II9-3	underground	intrusive					6.0	-0.25	0.05
II9-5	underground	intrusive					6.5	-0.24	0.02

Abbreviations: carb = carbonate minerals, Px = pyroxene.

[#] The analytical error is reported as twice the standard deviation from the mean.**Table 2**

Mg isotope data for marble xenolith and hybrid rock in the contact zone.

Sample	Rock type	Distance from the 1st sample (cm)	$\delta^{26}\text{Mg}$ (‰)	2SD
JC18-1	xenolith	0	-1.48	0.03
JC18-2	hybrid rock	0.8	-1.26	0.11
JC18-3	hybrid rock	1.3	-1.29	0.07
JC18-4	hybrid rock	2.2	-0.92	0.01
JC18-5	hybrid rock	2.9	-1.40	0.04
JC18-6	hybrid rock	4.2	-1.34	0.07
JC18-7	hybrid rock	4.9	-1.29	0.02
JC18-8	hybrid rock	6.8	-1.37	0.04
JC18-9	hybrid rock	7.7	-1.26	0.01
JC20-1	xenolith	0	-1.46	0.09
JC20-2	xenolith	2	-1.35	0.03
JC20-3	hybrid rock	4	-1.28	0.05
JC20-4	hybrid rock	6	-1.56	0.05
JC20-5	hybrid rock	8	-1.37	0.12
JC20-6	hybrid rock	10	-1.47	0.06
JC20-7	hybrid rock	12	-1.52	0.02
JC20-8	hybrid rock	14	-1.52	0.03
JC20-9	hybrid rock	16	-1.17	0.01
JC20-10	hybrid rock	18	-1.69	0.06

bracketing method against DSM-3: $\delta^{26}\text{Mg}$ (‰) = $[(^{26}\text{Mg}/^{24}\text{Mg})_{\text{sample}} / (^{26}\text{Mg}/^{24}\text{Mg})_{\text{DSM-3-1}}] \times 1000$. The long-term external precision was better than $\pm 0.05\text{‰}$ (2SD) based on analyses of in-house standards IGG ($-1.74 \pm 0.05\text{‰}$; 2SD, $n = 251$) and IEE ($0.312 \pm 0.05\text{‰}$; 2SD, $n = 210$). The $\delta^{26}\text{Mg}$ values for USGS standards BIR-1 ($-0.20 \pm 0.02\text{‰}$; 2SD, $n = 3$) and BCR-2 ($-0.18 \pm 0.04\text{‰}$; 2SD, $n = 3$) were consistent with literature values within error (An et al., 2014; Teng et al., 2010).

5. Results

5.1. Major element compositions

The major element compositions of marble, xenolith and hybrid rock, as well as ultramafic intrusive rocks (recalculated to 100% on a volatile-free basis) are plotted in Fig. 3, along with the estimated composition of parental magma (Li and Ripley, 2011). Results of xenoliths and hybrid rocks from Lehmann et al. (2007) are also plotted in Fig. 3. The contents of SiO_2 , Al_2O_3 and $\text{Fe}_2\text{O}_3^{\text{T}}$ of country rock marbles are low, and the content of MgO correlates with dolomite abundance. The compositions of the parental magma and ultramafic intrusive rocks plot along linear trends indicating the process of fractional accumulation. Xenoliths and hybrid rocks can be divided into three subgroups: one group following the trajectory from the compositions of marble, one group between the compositions of marble and intrusive rock, and the last group between the end of the marble trajectory and ultramafic intrusive rocks.

5.2. Mineral compositions

Dolomite is the main mineral in marble and exists in hybrid rocks of the contact zone. The MgO content of dolomite in marble varies from 21.5 wt% to 21.7 wt% and CaO content varies from 30.3 wt% to 30.5 wt% (Fig. 4a). Dolomite in hybrid rocks shows MgO contents from 20.1 wt% to 21.4 wt% and CaO contents from 29.8 wt% to 29.9 wt% (Fig. 4a). Dolomite in marble has low FeO contents (0.23 wt% to 0.47 wt%), and dolomite in hybrid rocks is characterized by higher FeO contents (0.58 wt% to 1.18 wt%). The MgO content of dolomite increases with CaO content (Fig. 4a), but decreases with FeO content (Fig. 4b). Sample (JC27) from the margin of the Jinchuan intrusion contains small dolomite grains, which have low CaO and MgO contents and the highest FeO^{T} contents.

Olivine in hybrid rocks is characterized by elevated Fo contents (Fo

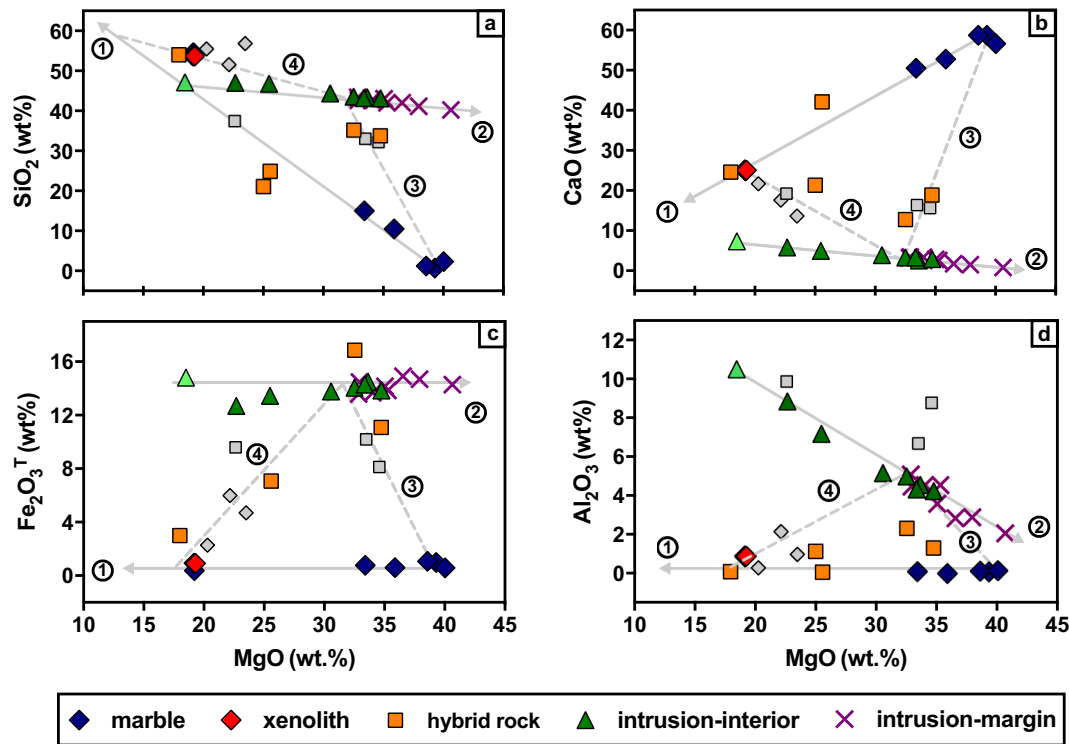


Fig. 3. Variations in major element compositions of rocks from marble, xenolith, hybrid rock and intrusive rock. The compositions are normalized to 100% on a volatile-free basis. The bright green triangle shows the composition of the parental liquid, as estimated by Li and Ripley (2011). The trajectory from marble is marked with ①, indicating the process of de-carbonation. The line ② starts from the parental liquid and passed through intrusive samples, showing the rocks are olivine cumulates. Lines ③ and ④ represent the mixing between mantle-derived magma and xenolith-xenolith derivatives.

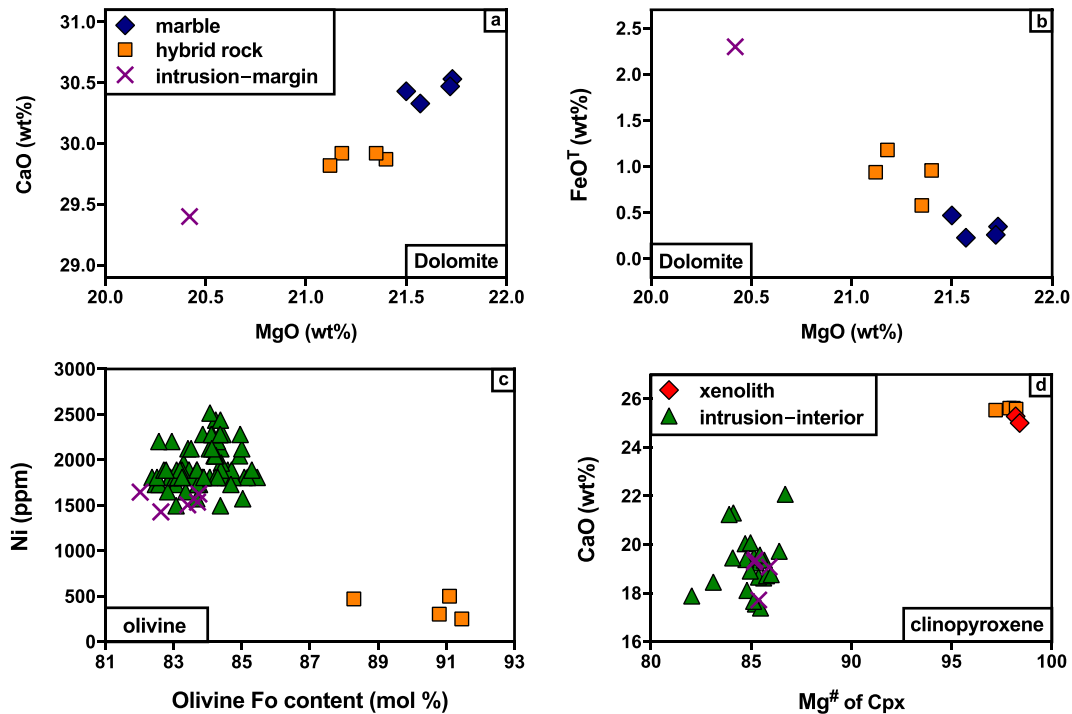


Fig. 4. Compositional variations of (a–b) dolomite, (c) olivine and (d) clinopyroxene. Data are from this study, [Chai and Naldrett \(1994\)](#) and [Li et al. \(2004\)](#).

$= 100 \times \text{Mg}/[\text{Mg}+\text{Fe}]$, molar in olivine) between 88 and 92, and low Ni contents between 250 and 501 ppm (Fig. 4c). In contrast, olivine in the interior and margin of the Jinchuan intrusion has Fo contents ranging from 82 to 86 and Ni contents from 1429 to 2517 ppm (Li et al., 2004;

this study). The Mg numbers of clinopyroxene ($\text{Mg}^\# = 100 \times \text{Mg}/[\text{Mg}+\text{Fe}^{\text{tot}}]$) in xenoliths and hybrid rocks vary from 97 to 98 (Fig. 4d), which are significantly higher than those for clinopyroxene from the interior and margin of the intrusion (82–87; [Chai and Naldrett, 1994](#);

this study). Clinopyroxene in xenoliths and hybrid rocks is characterized by CaO contents that range from 25 wt% to 26 wt% (Fig. 4d), whereas clinopyroxene in the interior of the intrusion has much lower CaO concentration, from 16 wt% to 22 wt% (Li et al., 2004).

5.3. C–O–Mg isotopic compositions

The $\delta^{18}\text{O}$ and $\delta^{13}\text{C}$ for carbonate minerals in marbles range from 13‰ to 22‰ and -1.7‰ to 1.3‰ , respectively (Fig. 5a). Carbonate minerals in xenoliths and hybrid rocks from the contact zone have lower values than those in marbles, with $\delta^{18}\text{O}$ and $\delta^{13}\text{C}$ values from 11‰ to 16‰ and -4.7‰ to -2.3‰ , respectively. In contrast, the variation of Mg isotope ratios in carbonate minerals is very small (Fig. 5b). The $\delta^{26}\text{Mg}$ values of marble range from -1.74‰ to -1.34‰ , and overlap the range of -1.71‰ to -1.36‰ in xenoliths and hybrid rocks.

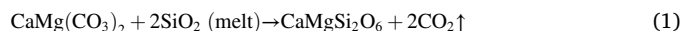
The $\delta^{18}\text{O}$ values of pyroxenes in the margin and interior of the Jinchuan intrusion range from 5.1‰ to 7.8‰ (Fig. 5c). Pyroxenes in the ultramafic intrusive rocks have $\delta^{26}\text{Mg}$ values between -0.32‰ and -0.16‰ (Fig. 5d), which are close to the estimated value of the Earth's mantle ($-0.25\text{‰} \pm 0.07\text{‰}$; Teng et al., 2010). Compared with the results of Tang et al. (2018), most samples are consistent within the error range (Supplementary Fig. S2).

Whole-rock magnesium isotopic compositions of hybrid rocks located at distances between 0 and 10 cm and 0 and 20 cm from two marble xenoliths are shown in Fig. 6a and b. The $\delta^{26}\text{Mg}$ values of hybrid rocks vary from -1.69‰ to -0.92‰ , with an average $\delta^{26}\text{Mg}$ value of $-1.38\text{‰} \pm 0.14\text{‰}$ ($n = 15$). The adjacent marble xenoliths have $\delta^{26}\text{Mg}$ values between -1.48‰ and -1.35‰ , which match well with the values of marble country rocks.

6. Discussion

6.1. Xenolith capture and associated reactions

Mineral assemblages in the contact zone reveal reactions between mafic magma and marble xenoliths as previously described by Lehmann et al. (2007) and Tonnelier (2010). The adjacent country rock is dominated by dolomite with small amounts of calcite. Marble xenoliths have been completely transformed into diopside + calcite. The silicate minerals in hybrid rocks at the contact zone are composed of olivine and diopside, and the carbonate minerals are calcite and dolomite. Reactions that produced silicate minerals were dependent on the activity of SiO_2 derived from magma, and can be described as:



In addition MgO may have been added to magma via reactions such as:

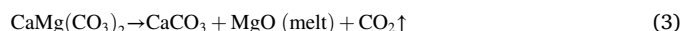


Fig. 7 illustrates the importance of $a\text{SiO}_2$ and $a\text{CO}_2$ at P-T conditions appropriate to the formation of olivine and diopside in the contact zone. The red line indicates the sequence of minerals and mineral reactions along the reaction path. At the conditions of low $a\text{SiO}_2$ and $a\text{CO}_2$, reaction (1) is stable and dolomite is converted into diopside. Once xenoliths were captured, reaction (1) progressed and CO_2 was generated. Formation of olivine by reaction (2) occurred over a restricted range of both $a\text{SiO}_2$ and $a\text{CO}_2$, where $\log a\text{CO}_2$ and $\log a\text{SiO}_2$ must lie between the dolomite-diopside-olivine-calcite invariant point and the dolomite-olivine-calcite-periclase invariant point (Fig. 7). Reaction (2) did not proceed until there was a sufficient supply of SiO_2 from the melt, or until significant degassing of CO_2 from reaction (1).

Xenoliths were captured in the contact zone and interacted with

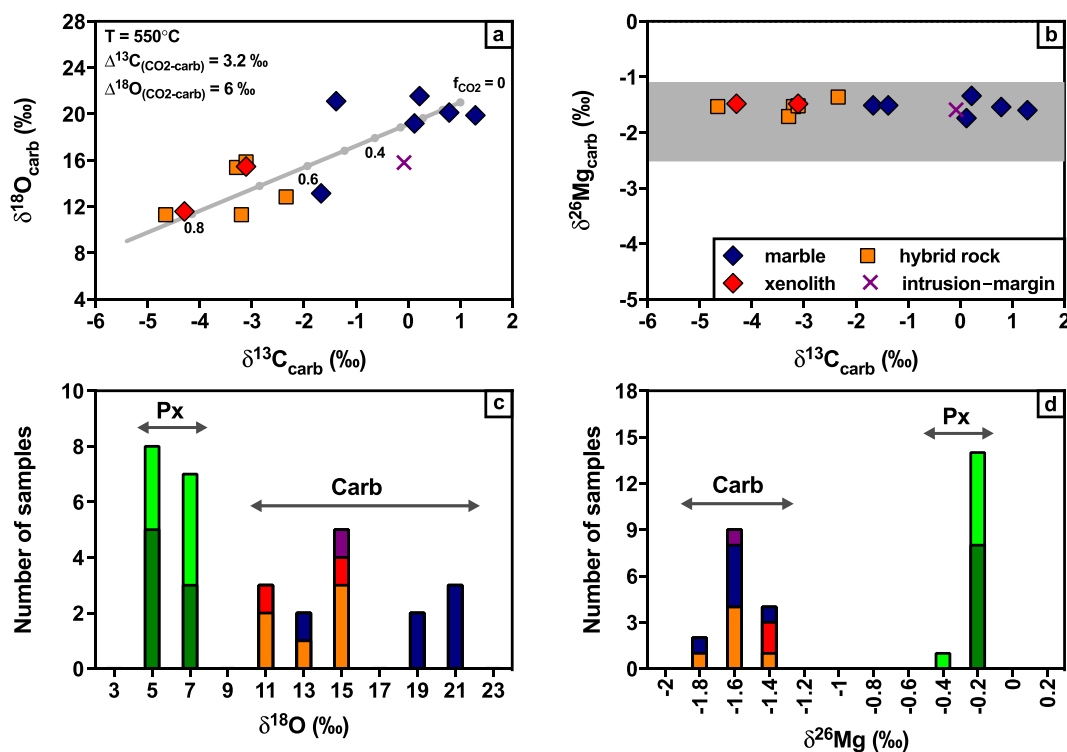


Fig. 5. (a) Variations of $\delta^{18}\text{O}$ values against $\delta^{13}\text{C}$ values for carbonate minerals. The released amounts of CO_2 (f_{CO_2}) are plotted along the trend of the Rayleigh fractionation model. (b) Variations of $\delta^{26}\text{Mg}$ values against $\delta^{13}\text{C}$ values for carbonate minerals. The gray shaded area represents the Mg isotopic compositional range of average mantle ($\delta^{26}\text{Mg} = -0.25\text{‰} \pm 0.07\text{‰}$; Teng et al., 2010). (c) and (d) Histograms of $\delta^{18}\text{O}$ values and $\delta^{26}\text{Mg}$ values for carbonate and pyroxene separates.

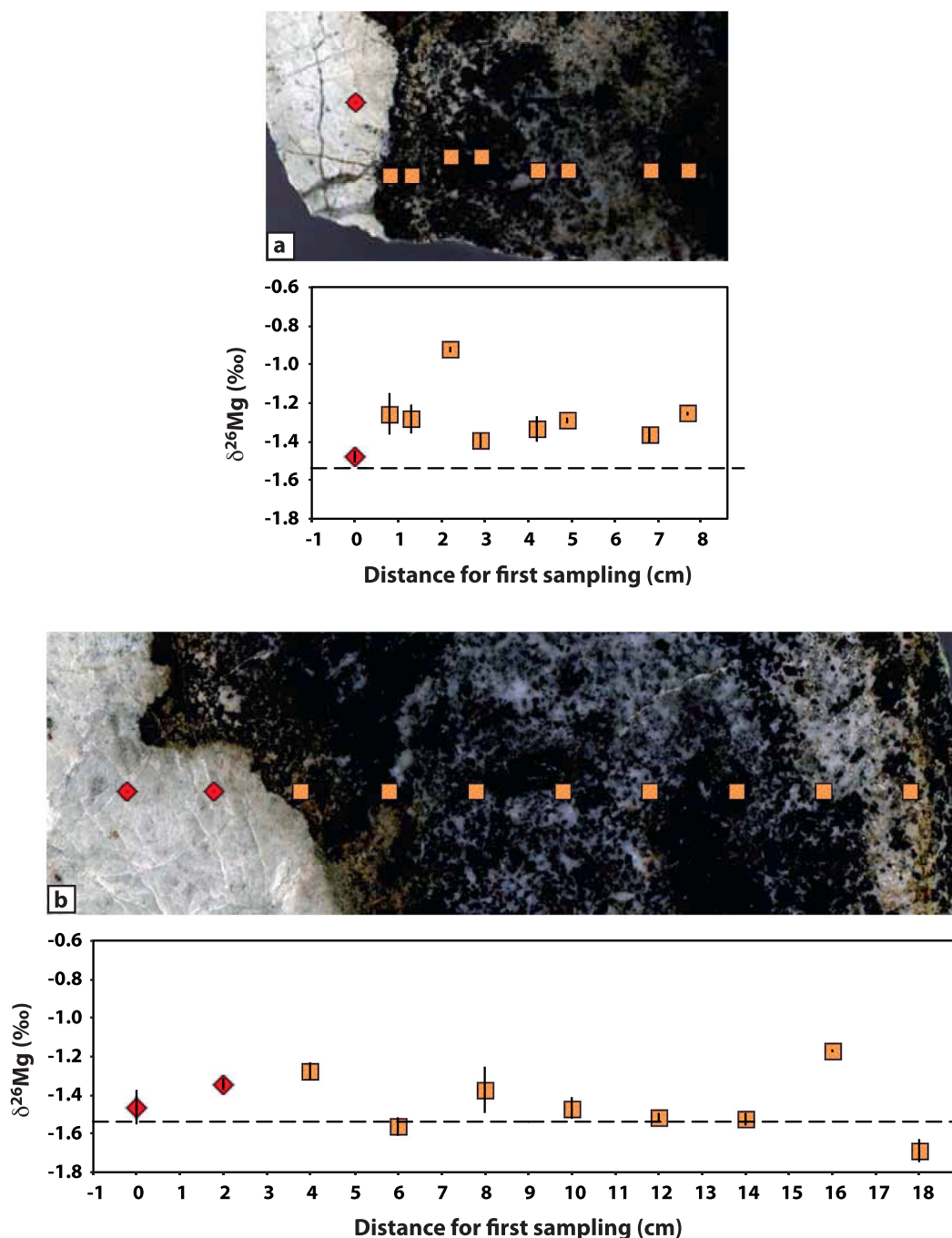


Fig. 6. Photographs of xenoliths and hybrid rocks (upper panel) illustrate the sample drilling sites. The Mg isotopic compositions of the corresponding samples are shown in the lower panels. The dashed lines represent the average $\delta^{26}\text{Mg}$ value of measured country rock marbles.

mantle-derived magma following the sequence of reactions (1) to (3). In addition to xenolith capture, the magma must lose heat to the surrounding country rock. However, the extremely low permeability (10^{-16} to 10^{-18} m²) and porosity (close to 0) of marble would result in inefficient heating, possibly slower than the rate of thermo-mechanical erosion (Huppert and Sparks, 1985; Williams et al., 1998). De Waal et al. (2004) suggested that olivine-bearing magma flowed through the Jinchuan conduit system. Tonnelier (2010) showed that part of the olivine crystallized in-situ, which in turn means that there must have been magma transportation. However, the timescale via thermal diffusion is estimated to have been around millions of years—two orders of magnitude longer than the time required for an intrusion to solidify by diffusion (Robertson et al., 2015). Even if a very high flow rate swept

away cooled magma and replaced it with new hot magma, we suggest that the incorporation of xenoliths is the dominant process and direct heating by magma likely had a negligible effect on reactions (1) to (3).

6.2. C-O isotopic evidence for de-carbonation

Carbonate minerals from xenoliths and hybrid rocks have lower carbon and oxygen isotopic ratios than those from marbles (Fig. 5a). As the carbon isotope fractionation between CO₂ and carbonate becomes positive at $T > 200$ °C (Bottinga, 1968; Horita, 2014), de-carbonation can effectively reduce $\delta^{13}\text{C}$ values of the remaining carbonate. Using a Rayleigh fractionation model, the integrated $\Delta^{13}\text{C}_{\text{CO}_2-\text{CC}}$ value was estimated to be $\sim 3.2\%$ at 550 °C. The $\delta^{18}\text{O}$ values are consistent with

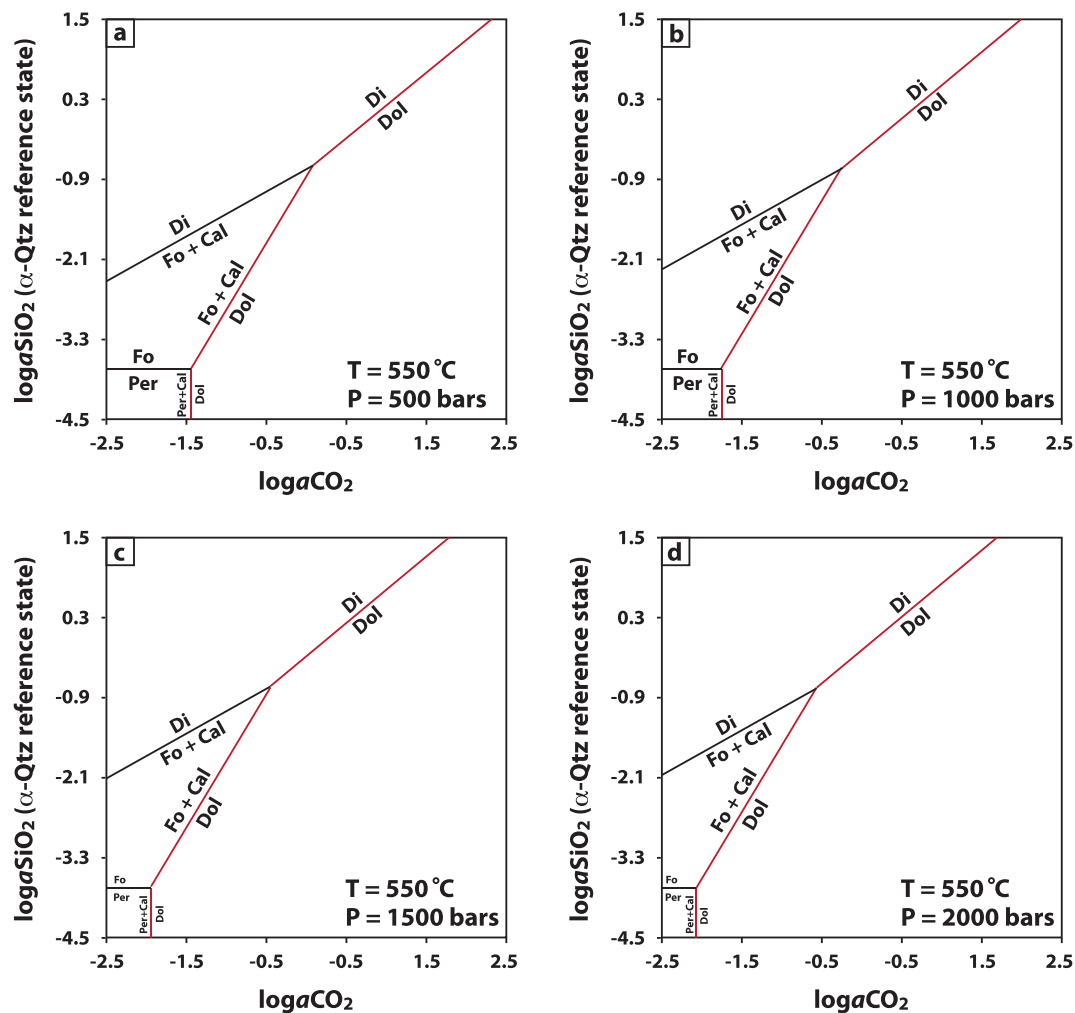


Fig. 7. Plot of mineral assemblages as a function of a_{SiO_2} and a_{CO_2} at a temperature of 550 °C and variable pressure from 500 to 2000 bars. The red line on each plot represents an individual reaction path. Thermodynamic calculations were performed using the database of [Holland and Powell \(2011\)](#) and the equation of state for $\text{CO}_2\text{-H}_2\text{O}$ fluid from [Kerrick and Jacobs \(1981\)](#). The calculations were performed within the software *Perple_X* 6.8.9 ([Connolly, 2005](#)).

those predicted assuming Rayleigh fractionation, and match an integrated $\Delta^{18}\text{O}_{\text{CO}_2\text{-CC}}$ value of $\sim 6\%$ at 550 °C ([Zheng, 1999](#)). [Ferry et al. \(2002\)](#) compared the contact metamorphism of siliceous carbonate rocks from 15 aureoles around the world and summarized peak temperatures for contact metamorphism in the range of 400 °C to 750 °C. [Nabelek \(2007\)](#) constructed a series of $T\text{-CO}_2\text{-}t$ paths for metamorphic reactions in carbonate-silicate contact aureoles; the separation of CO_2 occurs at temperatures of ~ 400 °C to 600 °C.

[Fig. 5a](#) illustrates the calculated $\delta^{18}\text{O}$ versus $\delta^{13}\text{C}$ depletion path for carbonate minerals collected from country rocks and contact zones. The $\delta^{13}\text{C}$ and $\delta^{18}\text{O}$ values are not only controlled by the temperature-dependent fractionation factor, but also by the amount of released CO_2 . Following the modeled Rayleigh fractionation line ([Fig. 5a](#)), calculations would require the extreme CO_2 loss from xenoliths and hybrid rocks ($f_{\text{CO}_2} = \sim 40\%$ to 80%). In contrast, carbonate minerals in adjacent country rocks experienced considerably less de-carbonation ($f_{\text{CO}_2} = 0\%$ to $\sim 40\%$). We suggest that the amount of CO_2 released from the contact zone was much more than that of adjacent country rock, which is consistent with the incorporation of xenoliths as the dominant process during magma-carbonate interaction at Jinchuan.

6.3. Mg isotope behavior in magma-carbonate interaction

Hybrid rocks in the contact zone contain olivine with elevated Fo content (>88) than the olivine in the intrusion (max Fo86). Diopside in

xenoliths and hybrid rocks is characterized by $\text{Mg}^\# > 97$, which is much higher than the $\text{Mg}^\#$ of clinopyroxene in ‘normal mafic rocks’ from the interior of the Jinchuan intrusion ([Fig. 4d](#)). It is likely that the MgO enrichment in olivine and diopside was related to the breakdown of dolomite via [reaction \(3\)](#).

Carbonate minerals from xenoliths and hybrid rocks have similar $\delta^{26}\text{Mg}$ values ranging from -1.71% to -1.36% ([Fig. 5b](#)), which are at the high end of published carbonate Mg isotopic ratios that range between -5.5% and -1% ([Galy et al., 2002](#); [Higgins and Schrag, 2010](#)). The similarity between the $\delta^{26}\text{Mg}$ values of carbonate minerals in contact zones and country rock marble ([Fig. 5b](#)) are consistent with country rocks being the source of captured xenoliths. The lack of variation in $\delta^{26}\text{Mg}$ values suggests that the conversion into calcite by [reactions \(2\) and \(3\)](#) did not change the Mg isotopic compositions, and that Mg isotope fractionation involving dolomite and calcite was not influenced by chemical and thermal gradients at contact zones.

Hybrid rocks located ~ 10 to ~ 20 cm from a xenolith margin show whole rock $\delta^{26}\text{Mg}$ values mostly in the range of marble country rock ([Fig. 6a](#) and [b](#)), which strongly suggests that Mg was largely sourced from marble xenoliths. Because the $\delta^{26}\text{Mg}$ values of the contact zone Mg differ from those of mantle-derived magma, the potential importance of Mg isotope fractionation between minerals should be considered. The isotopic differences between most mafic silicate mineral pairs are limited (e.g. $\Delta^{26}\text{Mg}_{\text{clinopyroxene-olivine}} = 0\%$ to 0.13% , $\Delta^{26}\text{Mg}_{\text{orthopyroxene-olivine}} = -0.04\%$ to 0.04% , [Handler et al., 2009](#); [Yang et al., 2009](#);

Young et al., 2009; Liu et al., 2011). Teng et al. (2010) observed no significant fractionation ($<0.1\%$) between minerals during differentiation of mafic magmas. Shen et al. (2013) pointed out no significant isotopic fractionation between minerals and granodioritic magma with normal, mantle-like isotopic signatures. Given the lack of Mg isotope fractionation between minerals and melts at high temperature, we suggest that the combined effects of dolomite break-down and the mixing of country rock-derived Mg and Mg from mafic magma were responsible for the significant difference of the $\delta^{26}\text{Mg}$ values of the contact zone relative to those of mantle-derived magma.

Indeed, the whole rock $\delta^{26}\text{Mg}$ value of hybrid rocks can be as high as -0.92% . The relative proportion of Mg from mantle-derived magma was calculated using a mass balance-mixing model based on volatile-free compositions. The Jinchuan parental magma is suggested to have formed by melting of metasomatized pyroxenitic mantle, likely produced by the interaction of recycled oceanic crust with variably-enriched peridotitic mantle (Tonnelier, 2010; Lu et al., 2019). However, arc lavas with heterogeneous Mg isotopic compositions are more likely to have been formed in subduction zones with depths deeper than 130 km and slab surface temperature hotter than 770°C (Hu et al., 2020). Therefore, we suggest that the mantle source of the magma that produced the Jinchuan intrusion had a mantle-like Mg isotopic composition similar to pyroxenite metasomatic samples from the Avacha volcano in the southern Kamchatka arc with $\delta^{26}\text{Mg}$ value from -0.30% to -0.21% (Li et al., 2017; Hu et al., 2020). Assuming a xenolith with an initial MgO content of 13 wt% and $\delta^{26}\text{Mg}$ value of -1.5% (generated via de-carbonation) reacted with mantle-derived magma having a MgO content of 11 wt% and a $\delta^{26}\text{Mg}$ value of -0.25% , our calculation shows that $\sim 52\%$ involvement of magma was necessary to reach the upper limit of the observed whole rock $\delta^{26}\text{Mg}$ values (Fig. 8). If marble country rock is taken as the end-member, the required amount of mantle-derived magma would be significantly increased (Fig. 8). Using a country rock marble with a MgO content of 40 wt% and a $\delta^{26}\text{Mg}$ value of -1.5% , $\sim 76\%$ contribution from magma would have been required to reach the value of -0.92% in the contact zone. If magma mixed with the MgO-melt released from reaction (3), the contamination end-member was composed of 100 wt% MgO with a $\delta^{26}\text{Mg}$ value of -1.5% . Calculated results show that the $\delta^{26}\text{Mg}$ value of whole rock would be -0.92% when the contribution of mantle-derived magma reached $\sim 89\%$ (Fig. 8). Depending on the choice of assimilated end-member, variable amounts of mantle-derived magma may have been involved in the assimilation process and resulted in local deviation of whole rock Mg isotopic values.

6.4. Reactive assimilation: chemical and isotopic trends

Reactions (1) to (3) involved the release of CO_2 and, for reactions 1 and 2, the conversion of dolomite to Mg-bearing silicate minerals. The C-

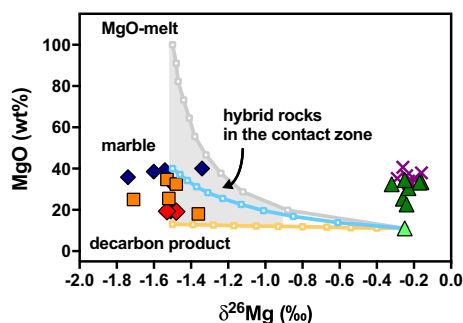


Fig. 8. Mixing model showing relative contribution of Mg from isotopically light xenolith-xenolith derivatives and isotopically heavy mantle-derived magma. The sample symbols are the same as those in Fig. 3. The gray shaded area represents the Mg isotopic compositional range of analyzed hybrid rocks.

O isotopes and Mg isotopes behaved differently during de-carbonation and dolomite breakdown. Fig. 9a and b show the correlation between whole rock CaO/SiO_2 ratios and $\delta^{18}\text{O}$ values of carbonate minerals, and between MgO/SiO_2 ratios and $\delta^{18}\text{O}$ values. For reaction (1) to reaction (3), the contribution of SiO_2 from mantle-derived magma correlates with the release of CO_2 and associated decrease in $\delta^{18}\text{O}$ values. However, the $\delta^{26}\text{Mg}$ values of carbonate minerals are constant even though the whole rock CaO/SiO_2 and MgO/SiO_2 ratios vary (Fig. 9c and d). There is no record of Mg isotope fractionation between reactant dolomite and produced Mg-bearing carbonate and silicate minerals in the contact zone.

Unlike carbon and oxygen that were capable of migrating out of the contact zone, magnesium reacted with SiO_2 and was fixed within a restricted spatial scale (Fig. 9f). Mantle-derived magma supplied SiO_2 that was required for chemical reactions, following the Si-gradient between magma and marble xenoliths. The local anomaly in $\delta^{26}\text{Mg}$ values in whole rocks is a function of the contribution of mantle Mg relative to that from xenoliths as described above. Elevated concentrations of MgO resulted in the formation of Mg-rich olivine and clinopyroxene enriched in the $\text{CaMgSi}_2\text{O}_6$ component. There is no evidence for the transport of MgO from the contact zone into the intrusion.

6.5. Implication for the genesis of the Jinchuan deposit

Lehmann et al. (2007) proposed that hot, primitive magma assimilated granitic material in the lower crust and partially crystallized. When the olivine-charged magma reached its present level in the middle to upper crust, country rocks were marbles and felsic gneisses. The assimilation of lower crustal wall rocks was characterized by input of a large amount of SiO_2 and Al_2O_3 (Song et al., 2012); this has been confirmed by evidence of mass-independent fractionation of S found in Jinchuan sulfides (Duan et al., 2016). Such assimilation of siliceous contaminants can cause a significant drop in the S concentration needed for the attainment of sulfide saturation (Ripley and Li, 2013). As fractional crystallization proceeded, sulfur was gradually enriched in residual magma. However, the residual magma would be too depleted in Ni if a high degree of olivine fractionation had occurred. Sulfide saturation at Jinchuan is therefore thought to have occurred during the early stage of fractional crystallization (Li and Ripley, 2011). It is likely that the incorporation of crustal sulfur is an essential element for early sulfide saturation (Duan et al., 2016). From the viewpoint of low sulfide solubility in silicate magma (Leshner, 2017, 2019), the large amount of sulfide that has accumulated in the Jinchuan deposit is difficult to explain by any process of in-situ sulfide segregation. Due to the existence of marble in the middle and upper crust, Lehmann et al. (2007) proposed another possibility that the assimilation of CO_2 derived from carbonate country rocks then increased the oxygen fugacity of magma and triggered the segregation of metal-rich sulfides at Jinchuan. In mafic silicate magma under moderately reducing conditions appropriate for mantle-derived magma, sulfide solubility is directly linked to the Fe^{2+} content (Haughton et al., 1974; Shima and Naldrett, 1975; Lehmann et al., 2007). An increase in oxygen fugacity will change Fe^{2+} to Fe^{3+} , reducing the effective Fe^{2+} content of magma and thus decreasing the sulfur concentration necessary to achieve sulfide liquid saturation.

The C-O isotopic compositions of xenoliths and hybrid rocks are consistent with the liberation of CO_2 as illustrated via reactions (1) to (3). The assimilation of a large amount of CO_2 may have caused the oxidation of mantle-derived magma, consistent with the oxidation state of the Jinchuan intrusion up to QFM +1.4 (Tonnelier, 2010). Compared with the Voisey's Bay deposit in Labrador, Canada, which represents the most reduced magmatic sulfide deposit known worldwide (QFM-1; Barnes et al., 2013), the magma that produced the Jinchuan deposit was more oxidized. The formation of magmatic deposits as a function of CO_2 assimilation is not rare. Ganino et al. (2014) proposed that the Agualblanca Ni-Cu deposit in Europe formed by contamination involving terrigenous sediments and carbonates. Ganino et al. (2008) and Tang

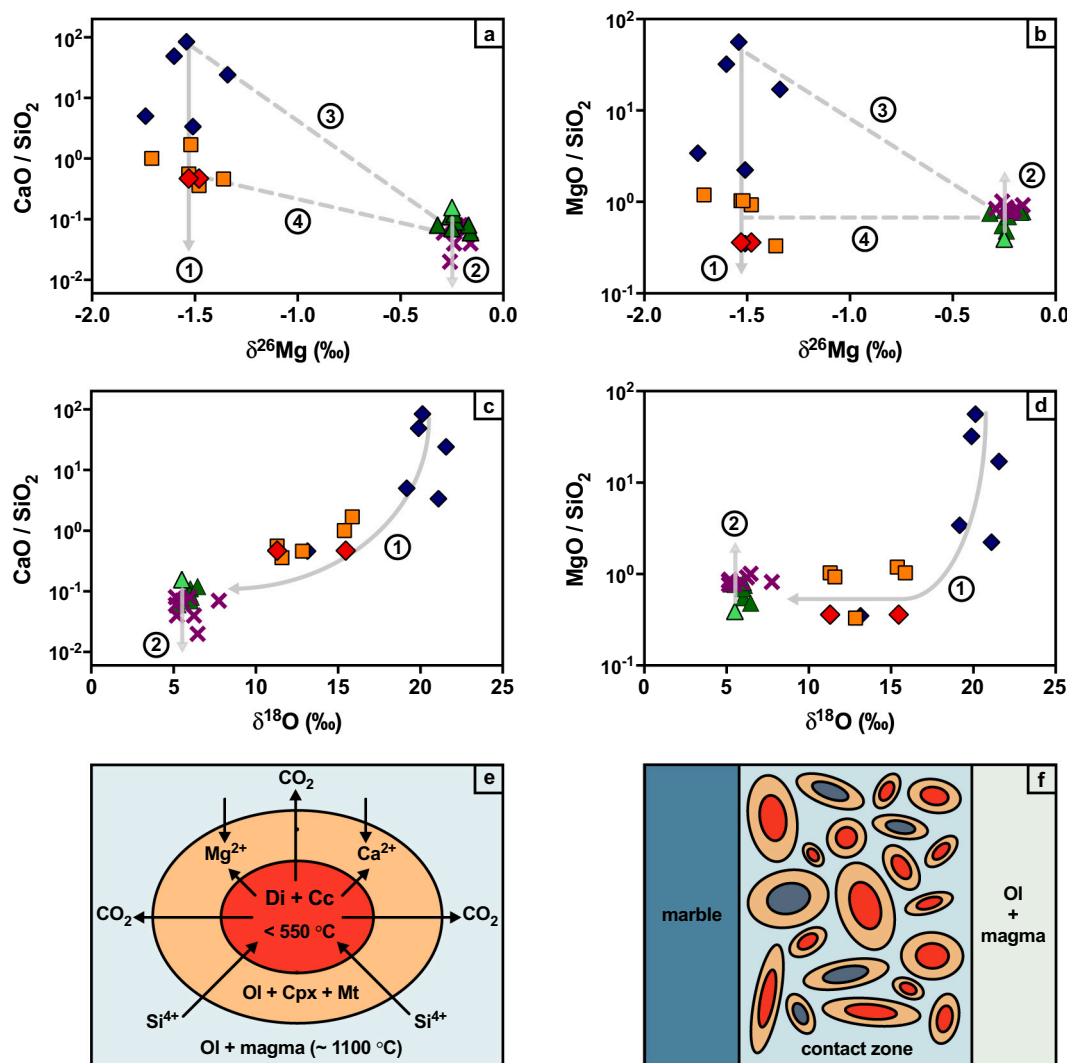


Fig. 9. Correlations between (a) $\delta^{26}\text{Mg}$ values and CaO/SiO_2 ratios, (b) $\delta^{26}\text{Mg}$ values and MgO/SiO_2 ratios, (c) $\delta^{18}\text{O}$ values and CaO/SiO_2 ratios, and (d) $\delta^{18}\text{O}$ values and MgO/SiO_2 ratios. The trajectory from marble is marked with , indicating the process of de-carbonation. The sample symbols are the same as those in Fig. 3. The line ② starts from the parental liquid and passed through intrusive samples, showing the rocks are olivine cumulates. Lines ③ and ④ represent the mixing between mantle-derived magma and xenolith-xenolith derivatives. (e) A model showing elemental exchange in the process of magma-carbonate interaction. (f) A cartoon illustrates the spatial constraints of chemical conversion and mechanical mixing. Except for carbon and oxygen, there is no evidence that other elements migrated out of the contact zone.

et al. (2017) proposed that the Panzihua-type Fe-Ti-V oxide deposits in the Emeishan large igneous province formed due to CO_2 assimilation by basaltic magma.

Mg isotopes appear not to be a good tracer of the ore-forming process in the Jinchuan mafic magmatic system. Heating by mantle-derived magma resulted in the partial melting and decarbonation of carbonate xenoliths, causing low-density, low-viscosity CO_2 to dissolve into the magma. There is no isotopic or chemical evidence that a MgO-rich melt moved away from the reactive assimilation front. Blythe et al. (2015) experimentally recorded the generation of CO_2 bubbles during magma-carbonate interaction and found that CO_2 bubbles can diffuse through the rheological barrier at melt interfaces. We suggest that reaction between xenoliths and the surrounding mantle-derived magma produced CO_2 bubbles and contaminated Mg-rich melt. Whereas CO_2 bubbles were efficiently transported away from the xenoliths, contaminated Mg-rich melt was confined to the contact zone. As illustrated in Fig. 3, the major element compositions of xenoliths and hybrid rocks are consistent with sustained volatile release, but confinement of Mg-rich melt to the proximity of xenoliths. This is consistent with the

pyroxene $\delta^{26}\text{Mg}$ values of ultramafic intrusive rocks being between -0.32‰ and -0.16‰ (Fig. 5d), which are close to the value of the Earth's mantle. This study reveals that the pyroxene $\delta^{26}\text{Mg}$ values at the margin and the center of the Jinchuan intrusion are similar (Fig. 9c and f), and do not show the involvement of carbonate-derived Mg. The ultramafic intrusive rocks at the margin of the Jinchuan intrusion can have pyroxene $\delta^{18}\text{O}$ values as high as 7.8‰ , consistent with contamination with xenolith-released CO_2 . Our data reveal that elevated oxygen isotope ratios of CO_2 were driven by de-carbonation and oxygen isotope exchange lead to ultramafic intrusive rocks with high $\delta^{18}\text{O}$ values. In contrast, magnesium isotope exchange involving Mg from country rocks and magma was restricted to the contact zone.

7. Conclusions

This study contrasts the behavior of Mg and C-O isotopes during magma-carbonate reactions at the Jinchuan Ni-Cu deposit. Xenolith mineral assemblages record a series of reactions that depend on the activities of SiO_2 and CO_2 to form olivine, diopside and MgO-melt. The

C-O isotopic compositions of country rocks and xenoliths show the evidence of de-carbonation with an integrated $\delta^{13}\text{C}_{\text{CO}_2-\text{CC}}$ value of $\sim 3.2\text{‰}$ and a $\delta^{18}\text{O}_{\text{CO}_2-\text{CC}}$ value of $\sim 6\text{‰}$, which are consistent with carbonate minerals releasing $\sim 40\%$ to 80% CO_2 at a temperature of approximately 550°C . However, Mg isotopes of carbonate minerals in the contact zone were not influenced by de-carbonation, showing $\delta^{26}\text{Mg}$ values similar to those of country rocks. The whole rock $\delta^{26}\text{Mg}$ values of xenoliths and hybrid samples vary from -1.69‰ to -0.92‰ , most of which are close to the value of marbles. The light Mg isotopic compositions can be explained by unbroken Mg-O bonds during the conversion from dolomite to Mg-bearing silicate minerals. The anomalous $\delta^{26}\text{Mg}$ value of -0.92‰ was related to local enrichment of mantle-derived magma in the contact zone. The $\delta^{26}\text{Mg}$ values of pyroxene from the margin and interior of the Jinchuan intrusion range from -0.32‰ to -0.16‰ , which are close to the value of the Earth's mantle.

Carbonate-magma interaction may be linked to the formation of the Jinchuan Ni-Cu sulfide deposit. The CO_2 released from de-carbonation reactions may have played a key role in increasing oxygen fugacity of magma, and ultimately the attainment of sulfide saturation. This study reveals that reactions involving dolomite breakdown and Mg liberation were restricted to the contact zone and that Mg was not transported into the intrusion. Without the input of country rock-derived MgO-melt, pyroxenes from the margin and interior ultramafic intrusive rocks show mantle $\delta^{26}\text{Mg}$ signatures.

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.chemgeo.2020.120044>.

Declaration of Competing Interest

None.

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